

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Geology and Geophysics

2. **Subject Name:** Hydrogeology    **L-T-P:** 3-0-0                      **Credits:** 3

3. **Pre-requisites:**

4. **Syllabus and reference books:**

### **Syllabus:**

The basic concepts of groundwater science and its relation with Earth systems; its impact on human and society; groundwater as a part of hydrosphere, hydrological equations, modeling of the hydrologic cycle, the water profile, sources of water, surface-groundwater interactions, recharge, and discharge processes, hydraulic parameters, and their influence on hydro dynamics groundwater flow, hydrogeochemistry, groundwater quality, types of groundwater, sources and pathways of groundwater solutes, groundwater pollution

### **Reference Books:**

1) Domenico, P.A. and Schwartz, F.W., 1998. *Physical and chemical hydrogeology*. New York: Wiley.

2) Fetter, C.W., 2018. *Applied hydrogeology*. Waveland Press

1) Tóth, J., 2009. *Gravitational systems of groundwater flow: theory, evaluation, utilization*. Cambridge University Press

2) Mukherjee, A., Scanlon, B.R., Aureli, A., Langan, S., Guo, H. and McKenzie, A.A. eds., 2020. *Global groundwater: source, scarcity, sustainability, security, and solutions*. Elsevier

5. **Lecture-wise break-up:**

| Sl. No.               | Topic                                                                                                                               | No. of lectures |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.                    | Concept of groundwater, its impact on human society                                                                                 | 4               |
| 2.                    | Global hydrologic processes and equations<br>Groundwater-surface water Interactions, state of systems, inflow and outflow processes | 8               |
| 3.                    | Type of water and Geological media                                                                                                  | 4               |
| 4.                    | Hydraulic properties of geologic media                                                                                              | 8               |
| 5.                    | Groundwater flow and hydrodynamics                                                                                                  | 6               |
| 6.                    | Groundwater Chemistry                                                                                                               | 9               |
| Total number of hours |                                                                                                                                     | 39              |